

ATHER SAYED

Software Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer who has shipped multiple live, production-deployed systems: an AI-powered restaurant management platform with XGBoost demand forecasting and Gemini LLM integration, an AI-powered hospital management platform (FastAPI, React, PostgreSQL, RAG/LLM features), and a real-time credit card fraud detection system combining XGBoost with an Isolation Forest anomaly signal. B.Sc. Electrical Engineering — Computer Science & Systems (Ain Shams University, 2024) with strong foundations in data structures, algorithms, OOP, and software design. Holds 13 AI/ML certifications from Stanford, DeepLearning.AI, and IBM, spanning deep learning, computer vision, NLP, and LLMs.

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, SQL, HTML5, CSS3

Web Development

React, Vite, Tailwind CSS, shadcn/ui, REST APIs, Responsive Design, Node.js (basics)

Backend & MLOps

FastAPI, REST APIs, Docker, Docker Compose, PostgreSQL, SQL, Git, GitHub, Linux, CI/CD, Model Deployment

Software Engineering

Object-Oriented Programming, Data Structures & Algorithms, Software Design, Testing, Code Review, Agile/Scrum

Machine Learning

Supervised & Unsupervised Learning, Deep Learning, CNNs, Transfer Learning, Reinforcement Learning, XGBoost, Anomaly Detection, Model Evaluation, Hyperparameter Tuning

Foundations

Linear Algebra, Calculus, Probability & Statistics, Data Preprocessing, Feature Engineering

Computer Vision

Image Classification, Medical Imaging, OpenCV, Data Augmentation, ONNX Inference

Libraries

TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Hugging Face

Generative AI & NLP

Large Language Models (LLMs), Generative AI, Transformer Architecture, Foundational Models, NLP, RAG (FAISS)

EXPERIENCE

Independent Technical Specialization — Machine Learning & Full-Stack Development

Dec 2025 – Present

Cairo, Egypt

- Pursued a full-time, intensive specialization in Machine Learning and Deep Learning immediately after completing mandatory military service in December 2025, earning 13 certifications from Stanford, DeepLearning.AI, and IBM.
- Designed, built, and deployed multiple full-stack and machine learning projects end-to-end — from data pipelines and model training to containerized production deployment (see Projects).

PROJECTS

AI-Powered Hospital Management System — Python, FastAPI, React, TypeScript, PostgreSQL, Redis, Docker, FAISS | [GitHub](#) | [Live Demo](#)

- Built a full-stack, AI-powered hospital management platform with JWT auth, Google OAuth, and role-based access control across five roles (admin, doctor, nurse, receptionist, patient).
- Implemented core hospital modules — patients, doctors, appointment scheduling with conflict detection, EMR with audit trail, and line-item billing.
- Developed AI features: multi-turn triage chatbot, medical report summarization, structured clinical notes, and a RAG-based clinical search pipeline (FAISS + LLM) with OCR document ingestion.
- Containerized the full stack with Docker Compose and deployed to production (Vercel, Railway, Supabase).

PizzaStore — AI-Powered Restaurant Management System — *FastAPI, React, TypeScript, PostgreSQL, XGBoost, Gemini API, Docker* | [GitHub](#) | [Live Demo](#)

- Built a full-stack restaurant management platform with JWT authentication and role-based access control across three roles (Admin, Manager, Cashier), deployed live on Railway and Vercel with Supabase PostgreSQL.
- Trained an XGBoost demand forecasting model on 48,000 real restaurant orders with RandomizedSearchCV hyperparameter tuning, achieving MAE of 1.74 units; implemented recursive 7-day multi-step forecasting with lag features and rolling averages.
- Built a natural-language query interface using Google Gemini, constrained to pre-written SQL templates with three independent security layers including a read-only PostgreSQL user at the database level.

FraudWatch AI — Real-Time Credit Card Fraud Detection — *Python, FastAPI, XGBoost, ONNX Runtime, React, TypeScript, Docker* | [GitHub](#) | [Live Demo](#)

- Designed and shipped an end-to-end fraud detection platform on the ULB/Kaggle credit-card dataset (~0.17% fraud rate), using a time-based train/test split to avoid data leakage.
- Trained an XGBoost model with class weighting as the primary signal, paired with an Isolation Forest as a secondary anomaly signal; served both via ONNX Runtime for low-latency inference.
- Tuned the decision threshold to ~75% recall / ~85% precision at 0.97, prioritizing catching real fraud over minimizing analyst review volume; built a FastAPI backend and React dashboard for live scoring and incident triage.

Game Hub — Game Discovery Web App — *React, TypeScript, Vite, Tailwind CSS, REST APIs* | [GitHub](#) | [Live Demo](#)

- Built a responsive, production-deployed web app for discovering games, with dynamic filtering, search, sorting, and platform/genre browsing.
- Architected reusable, type-safe React components and custom hooks in TypeScript, integrating a third-party REST API with loading states and error handling.

EDUCATION

Nozha Language Schools <i>Cairo, Egypt K-12 education</i>	2005 – 2019
Bachelor of Science in Electrical Engineering — Computer Science & Systems <i>Ain Shams University, Cairo, Egypt Very Good</i> <i>Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Design, Operating Systems, Computer Networks, Database Systems, Digital Signal Processing</i>	2019 – Sep 2024
National Military Service <i>Egyptian Armed Forces Completed mandatory national service</i>	Dec 2024 – Dec 2025

CERTIFICATIONS

- **Stanford & DeepLearning.AI** — Machine Learning Specialization: Supervised ML, Advanced Learning Algorithms, Unsupervised Learning/Recommenders/RL (3 courses) | 12/2025 – 01/2026
- **DeepLearning.AI** — Neural Networks and Deep Learning | 02/2026
- **IBM** — Machine Learning & Deep Learning Foundations: Machine Learning with Python, Neural Networks with Keras, Deep Learning with Keras/TensorFlow, Neural Networks with PyTorch, Deep Learning with PyTorch (5 courses) | 03/2026 – 04/2026
- **IBM** — Generative AI & Capstone: Architecture & Data Preparation, Foundational Models for NLP, AI Capstone Project (3 courses) | 05/2026 – 06/2026
- **CodeKloud** — Docker Training Course for the Absolute Beginner | 04/2026

LANGUAGES

Arabic (Native) English (Professional Proficiency)